

Production Readiness Note

Overview:

This project deploys a highly available and scalable **2-Tier AWS Infrastructure** using Terraform modules. The infrastructure includes VPC, Public & Private Subnets, Security Groups, ALB, ACM, Route53, Auto Scaling Group, IAM Roles, and CloudWatch Monitoring.

Production Ready Features

Multi-AZ Deployment for High Availability

Application Load Balancer for Traffic Distribution

Auto Scaling Group for Scalability

Private EC2 Instances (No Direct Internet Access)

HTTPS Enabled using ACM Certificates

Route53 DNS Integration

Security Group Based Access Control

IAM Role Based Permissions

CloudWatch Monitoring and Alarms

Infrastructure as Code using Terraform

Security Measures

- Private application servers deployed in private subnets.
- HTTPS enforced using ACM certificates.
- Least privilege IAM access.
- Security Group isolation between ALB and EC2.
- Encrypted EBS volumes.

Scaling Strategy

- Auto Scaling Group with Min: 2, Desired: 2, Max: 4 instances.
- ALB distributes traffic across healthy instances.
- Infrastructure can be scaled horizontally by increasing ASG capacity.

Future Improvements

- AWS WAF Integration
- AWS Backup
- AWS Secrets Manager
- SNS Notifications
- VPC Flow Logs

- RDS/Aurora Database Layer

Conclusion

The solution is production-ready for small to medium-scale workloads and follows AWS best practices for security, availability, and scalability. Additional enhancements can further strengthen operational and security capabilities.